

» Gradient Descent

- * Recall general iterative minimisation algorithm:

```
x=x0
for k in range(num_iters):
    step = calcStep(fn,x)
```

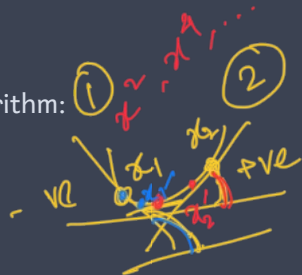
→ $x = x - \text{step}$

- * We know one way to choose the step, namely:

→
$$\text{step} = \alpha \left[\frac{\partial f}{\partial x_1}(x), \frac{\partial f}{\partial x_2}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right]$$

where α is the *step size* or *learning rate*

- * How to choose α ?

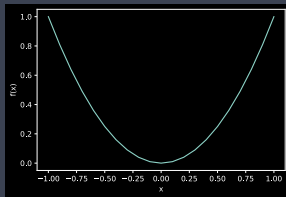


$$x = x - \alpha \nabla f$$

$$x - \alpha \nabla f = x - \alpha \frac{\partial f}{\partial x}$$

» Example

- * Minimise function $f(x) = x^2$. Derivative $df/dx = 2x$.



- * Minimum of $f(x) = 0$ when $x = 0$

- * Python code:

```
class QuadraticFn():  
    def f(self, x):  
        return x**2 # function value f(x)  
  
    def df(self, x):  
        return 2*x # derivative of f(x)  
  
def gradDescent(fn,x0,alpha=0.15,num_iters=50):  
    x=x0;  
    X=np.array([x]); F=np.array(fn.f(x));  
    for k in range(num_iters):  
        step = alpha*fn.df(x) ✓  
        x = x - step ✓  
        X=np.append(X,[x],axis=0); F=np.append(F,fn.f(x))  
    return (X,F)
```

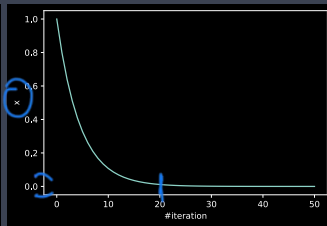
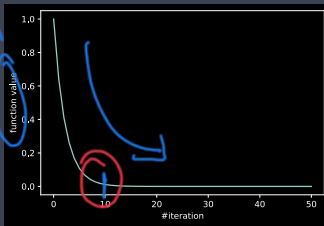
(X, F) = gradDescent(fn,x0=1,alpha=0.1)

Handwritten notes on the right side of the slide:

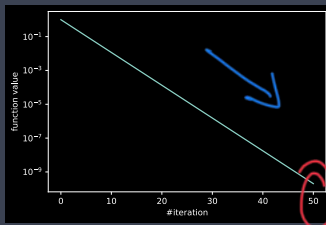
- x^2
- $k \in \mathbb{N}$
- $k \in \mathbb{Z}$
- $-x^2, x^2, x^4, 3x^2+5$

» Example

- * Starting value $x_0 = 1$, step size $\alpha = 0.1$



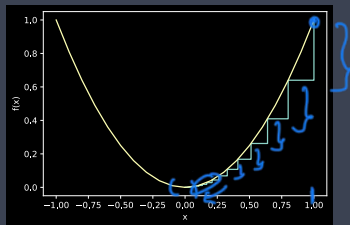
- * Function decreases at every iteration, decrease gets smaller over time. Input x also decreases from 1 towards 0 (the optimum value).
- * Hard to see small values of function, helps to plot on a log scale:



- * On log-scale the function value is decreasing linearly towards zero i.e. $\log f(x_t) = \underline{mt} + c$ with $\underline{m} < 0$, so $f(x_t) = e^{\underline{m}t+c}$ decreases exponentially $\rightarrow$ super fast!. This is basically best-case performance

» Example

- * Starting value $x_0 = 1$, step size $\alpha = 0.1$
- * Can be helpful to plot $(x, f(x))$ pairs of values over time.

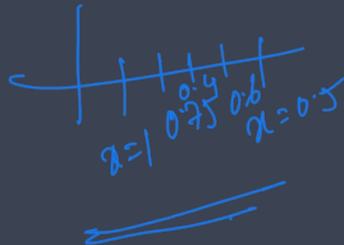
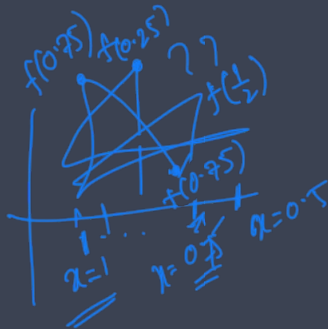
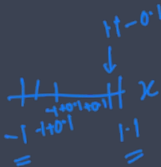


- * Why do the steps in x get smaller over time? Recall derivative is $2x$ and minimum is at $x = 0$. Derivates don't always diminish as get close to optimum, more on this later.

» Example

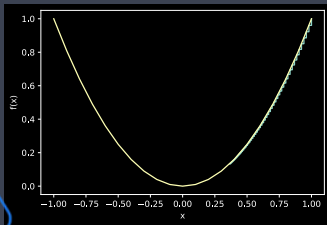
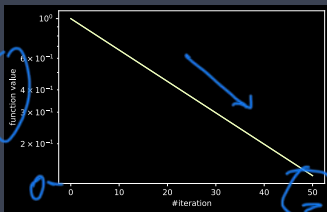
Plotting commands:

```
→ xx=np.arange(-1,1,1,0.1)
→ plt.plot(xx,fn.f(xx))
plt.xlabel('x'); plt.ylabel('f(x)')
plt.show()
plt.plot(F)
plt.xlabel('#iteration'); plt.ylabel('function value')
plt.show()
plt.semilogy(F)
plt.xlabel('#iteration'); plt.ylabel('function value')
plt.show()
plt.plot(X)
plt.xlabel('#iteration'); plt.ylabel('x')
plt.show()
plt.step(X,fn.f(X))
xx=np.arange(-1,1,1,0.1)
plt.plot(xx,fn.f(xx))
plt.xlabel('x'); plt.ylabel('f(x)')
plt.show()
```



» Example

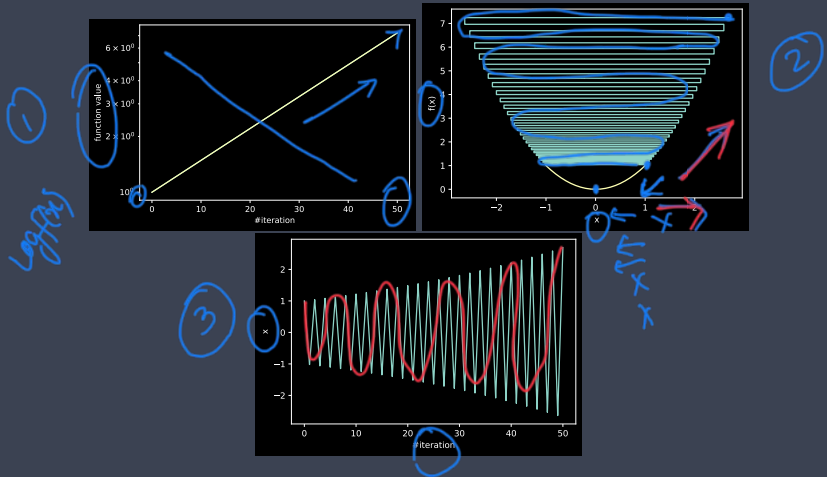
- * Starting value $x_0 = 1$, step size $\alpha = 0.01$



- * Now rate of convergence is much smaller, and steps in x are smaller too.

» Example

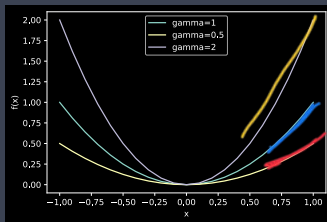
- * Starting value $x_0 = 1$, step size $\alpha = 1.01$



- * Now no longer converges to minimum (in fact it diverges), steps in x are larger.

» Example

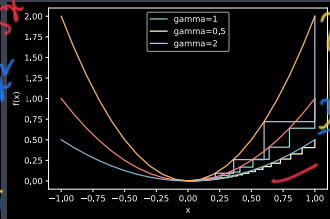
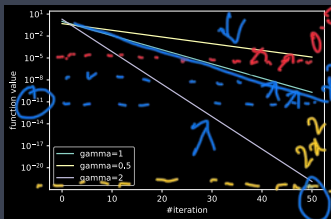
- * What happens as function becomes more curved? Try $f(x) = \gamma x^2$ and vary γ .



Handwritten notes:

- $2x^2$
- $x^2 \propto$
- $0.5x^2$
- $\gamma = 1 \quad x^2$
- $\gamma = \frac{1}{2} \quad 0.5x^2$
- $\gamma = 2 \quad 2x^2$
- $\frac{d}{dx} \gamma x^2 = 2\gamma x$

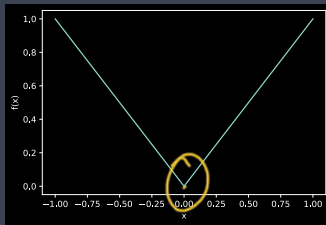
- * Starting value $x_0 = 1$, step size $\alpha = 0.1$



- * Derivative is $2\gamma x$, so steps in x increase/decrease as γ increases/decreases $\rightarrow$ so we can expect to have to adjust step size α based on magnitude of derivative of $f(\cdot)$.

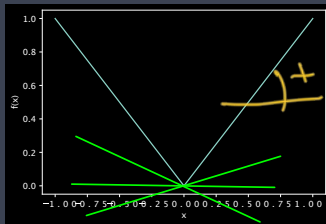
» Example 2

- * Another example: $f(x) = |x|$. Minimum is 0 when $x = 0$.



$$f(x) = |x| = \begin{cases} x & : x > 0 \\ 0 & : x = 0 \\ -x & : x < 0 \end{cases}$$

- * Slope is constant, it doesn't decrease as get closer to minimum
- * When $x > 0$ derivative of $f(\cdot)$ is $+1$, when $x < 0$ derivative is -1 i.e. derivative is $\text{sign}(x)$ for $x \neq 0$. When $x = 0$? No single line just touches the curve at that point ...

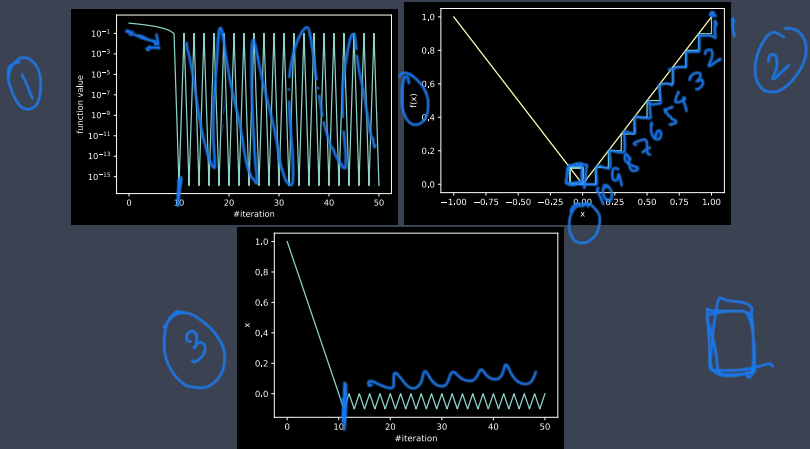


$$\frac{\partial f(x)}{\partial x} = \begin{cases} 1 & : x > 0 \\ -1 & : x < 0 \end{cases}$$

- * Define derivative at $x = 0$ to be 0, will come back to this later.

» Example 2

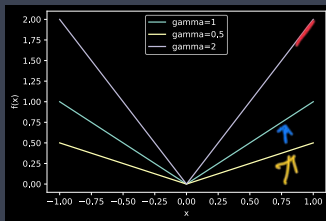
- * Starting value $x_0 = 1$, step size $\alpha = 0.1$



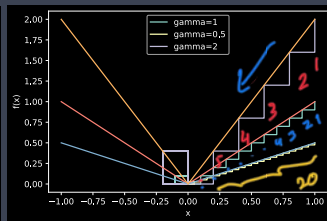
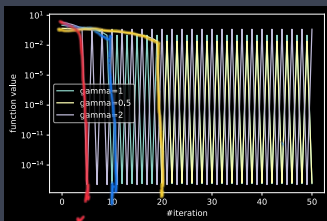
- * Function decreases until it reaches vicinity of minimum, then oscillates forever - never converges to minimum.
- * How might we try to fix that?

» Example 2

- * As usual, changing magnitude of derivative affects steps. $f(x) = |\gamma x|$, derivative is $\gamma \text{sign}(x)$ for $x \neq 0$.

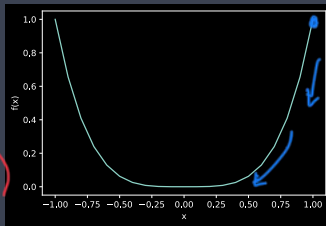


- * Starting value $x_0 = 1$, step size $\alpha = 0.1$

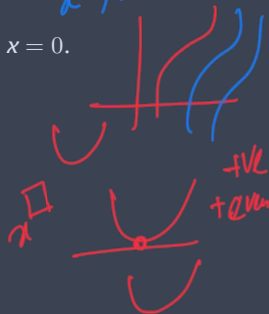
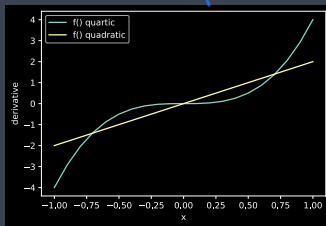


» Example 3

* Yet another example: $f(x) = x^4$. Minimum is 0 when $x = 0$.



* Slope decreases more quickly than quadratic as get closer to minimum. Derivative of $f(x) = x^4$ is $4x^3$. Compare with derivative $2x$ when $f(x) = x^2$.



$$4x^3$$

$$2x$$

$$y = mx + c$$

» Example 3

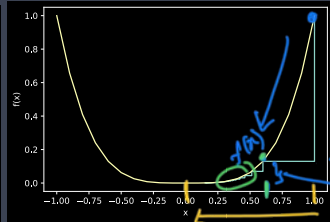
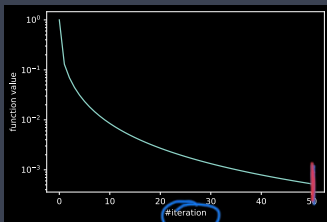
Δx

* Starting value $x_0 = 1$, step size $\alpha = 0.1$

Δx^3
 $x=1$ ✓
of

①

$f(x)$

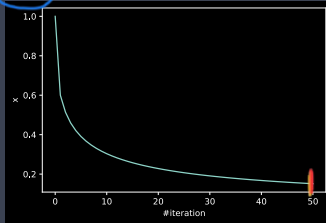


②

→ step 1

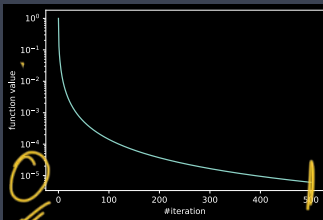
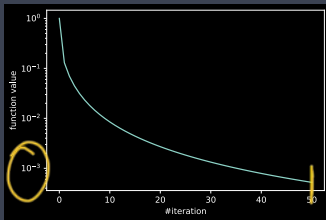
→ step 2

③

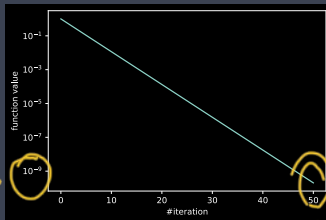


» Example 3

- * Starting value $x_0 = 1$, step size $\alpha = 0.1$



- * Observe that rate of decrease of $f(\cdot)$ gets slower and slower over time.
- * Decrease of $\log f(x)$ is no longer linear, convergence rate is *much* slower than for quadratic function:



- * How might we try to fix this?

» Summary

What have we discovered?

- * No “one size fits all” choice of step size.
 - * Scaling function e.g. from x^2 to $2x^2$ changes magnitude of derivative and so of steps taken
- * Maybe need to change the step size over time
 - * If derivative doesn't decrease as get close to minimum then can end up “bouncing around”
 - * If derivative decreases too quickly as get close to minimum then can end up with v slow convergence
- * Convergence rate can be v sensitive to properties of function being minimised
 - * Quadratic function is “best” case, exponentially fast convergence (jargon: quadratic-like → *strongly convex*)
 - * Kinks and flat areas in functions can greatly slow convergence (jargon: has kinks → *non-smooth*)

